

YOUR NAME
YOUR EMAIL
October 6, 2026

Homework 3

Don't forget an academic integrity statement (see syllabus for more details)!

1. Finish problems 2–4, then complete the academic integrity statement.
2. **Building off of the previous homework** In Homework 2, you should have performed an ANOVA on the dataset in the `rats.csv` file. If done correctly, you would have found a statistically significant treatment effect (at the $\alpha = 0.05$ level), with a P-value of 0.0102. That is, there is *some* evidence of differences between the treatment means.

We now would like to expand on these results.

- (a) What is the estimated treatment mean μ_i for each of the treatments?
 - (b) If you wanted to compare all treatment means against each-other (all pairwise comparisons), how many comparisons would you be making?
 - (c) Using an $\alpha = 0.05$ significance level, perform all pair-wise comparisons using Tukey's-HSD adjustment, and list the adjusted P-value for each comparison. What are the significant differences you found?
 - (d) **(For 6000-level students):** Now perform the same pair-wise comparisons without making the Tukey adjustment. Does it change the outcomes?
3. **This problem uses the TODO dataset in Canvas.** Twelve orange pulp silage samples were divided at random into four groups of three. One of the groups was left as an untreated control, while the other three groups were treated with formic acid, beet pulp, and sodium chloride, respectively. The response variable recorded in the file TODO is the moisture content of the silage.
 - (a) Compute an analysis of variance table for these data, and test the null hypothesis that all four treatments yield the same average moisture contents with a significance level of $\alpha = 0.05$.
 - (b) Perform all pair-wise comparisons using Tukey's-HSD adjustment method, and report the outcome of each pairwise comparison.
 - (c) **(For 6000-level students):** Using an appropriate contrast, compute a 99% confidence interval for the difference in response between the average of the three treatment groups (acid, pulp, and salt) and the control group.

4.